

7(a)	Work done to move the masses from infinity to the distance of separation/to that point.
(b)(i)	A geostationary orbit is one where the satellite will remain in the same position in the sky relative to the Earth's surface
(b)(ii)	The orbital period of the moon is not 24 hours
(b)(iii)	$T = 27.3 \times 24 \times 60 \times 60 = 2358720 \text{ s}$
	$\omega = \frac{2\pi}{T} = \frac{2\pi}{2358720} = 2.66 \times 10^{-6} \text{ rad s}^{-1}$
(b)(iv)	The gravitational force by the Earth on the Moon provides for the centripetal force.
	$\frac{GM_{\text{Earth}}M_{\text{Moon}}}{r^2} = M_{\text{Moon}}a_{\text{Moon}}$ $\frac{GM_{\text{Earth}}}{r^2} = r\omega^2$ $M_{\text{Earth}} = \frac{r^3\omega^2}{G}$
	$M_{\text{Earth}} = \frac{r^3\omega^2}{G}$ $M_{\text{Earth}} = \frac{(3.84 \times 10^8)^3 \times (2.6638 \times 10^{-6})^2}{(6.67 \times 10^{-11})}$ $M_{\text{Earth}} = 6.02 \times 10^{24} \text{ kg}$
(b)(v)1	The gravitational force by the Earth on the Moon provides for the centripetal force. $\frac{GM_e M_m}{x^2} = M_m \frac{v^2}{x}$ $\frac{GM_e M_m}{2x} = \frac{1}{2} M_m v^2$ Kinetic energy = $\frac{GM_e M_m}{2x}$
(b)(v)2	Gravitational potential energy = $-\frac{GM_e M_m}{x}$ Total energy = $-\frac{GM_e M_m}{x} + \frac{GM_e M_m}{2x}$ Total energy = $-\frac{GM_e M_m}{2x}$
(b)(v)3	When the total energy is zero (or more), this implies the Moon is able to escape from the Earth's gravitational field.
(b)(vi)1	As the Moon moves further from the Earth, its orbital radius/x increases
	The total energy increases.
	The potential energy will increase more than the decrease in kinetic energy
(b)(vi)2	Since this is an Earth-Moon system, the Earth loses energy while the Moon gains energy.
(b)(vi)3	As the Earth loses energy, the orbital speed about the Earth axis decreases. Since angular velocity is smaller, the period will be longer.
(b)(vi)	The position where the maximum gravitational potential is between M and N
	The direction of the gravitational field strength at N will be towards the moon

	It will accelerate towards the Moon instead.
--	--